A **perpetual bond** is a less common type of coupon bond with no stated maturity date. Most perpetual bonds are issued by companies to obtain equity-like financing and often include redemption features. As $N \to \infty$ in Equation 6, we can simplify this to solve for the present value of a **perpetuity** (or perpetual fixed periodic cash flow without early redemption), where $r > 0$, as follows:

$$PV(\text{Perpetual Bond}) = PMT / r. \quad (7)$$

EXAMPLE 4

KB Financial Perpetual Bond

In 2020, KB Financial (the holding company for Kookmin Bank) issued KRW325 billion in perpetual bonds with a 3.30 percent quarterly coupon.

1. Calculate the bond's YTM if the market price was KRW97.03 (per KRW100).

   Solution:

   3.40 percent
   Solve for r in Equation 7 given a PV of KRW97.03 and a periodic quarterly coupon (PMT) of KRW0.825 (= (3.30% × KRW100)/4):

   KRW97.03 = KRW0.825 / r ;

   r = 0.85% per period, or 3.40% (=0.85% × 4) on an annualized basis.

Annuity Instruments

Examples of fixed-income instruments with level payments, which combine interest and principal cash flows through maturity, include fully amortizing loans such as mortgages and a fixed-income stream of periodic cash inflows over a finite period known as an annuity. Exhibit 3 illustrates an example of level monthly cash flows based upon a mortgage.

Exhibit 3: Mortgage Cash Flows